

Seaborn Essential Cheat Sheet

Purpose: Quick reference for essential Seaborn plots used in EDA and ML preparation.

1. Basic Setup

```
import seaborn as sns
import matplotlib.pyplot as plt
```

General pattern: `sns.some_plot(data=df, x='column', y='column')`

2. Scatter Plot — Relationship

```
sns.scatterplot(data=df, x='Price', y='Quantity')
sns.scatterplot(data=df, x='Price', y='Quantity', hue='Category')
plt.title('Price vs Quantity')
plt.show()
```

Use: Relationship between two numerical variables.

3. Line Plot — Trend

```
sns.lineplot(x=days, y=sales, marker='o')
sns.lineplot(data=df, x='Day', y='Sales', marker='o')
plt.title('Daily Sales')
plt.show()
```

Use: Trends/change across an ordered variable, especially time.

4. Bar Plot — Category Comparison

```
sns.barplot(data=df, x='Category', y='Price')
plt.title('Average Price by Category')
plt.show()
```

Important: `barplot()` aggregates values; the default estimator is the mean.

5. Histogram — Distribution

```
sns.histplot(data=df, x='Price')
sns.histplot(data=df, x='Price', kde=True)
plt.show()
```

Use: See how numerical values are distributed. `kde=True` adds a smooth density estimate.

6. Box Plot — Spread & Outliers

```
sns.boxplot(data=df, y='Price')
sns.boxplot(data=df, x='Category', y='Price')
plt.show()
```

Use: Compare distributions, spread, median, and possible outliers.

7. Heatmap — Correlation

```
correlation = df[['Price', 'Quantity', 'Revenue']].corr()

sns.heatmap(correlation, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

.corr() calculates pairwise correlation between numerical columns. Values range from -1 to +1.

annot=True → writes values inside cells.

cmap='coolwarm' → controls colors.

Interpretation: +1 = strong positive, 0 = little/no linear relationship, -1 = strong negative. **Correlation does not imply causation.**

8. Pair Plot — Many Relationships

```
sns.pairplot(
df,
vars=['Price', 'Quantity', 'Revenue'],
hue='Category'
)
plt.show()
```

Use: Quick visual overview of relationships among several numerical variables. It creates a grid of scatter plots and distributions.

9. Plot Decision Cheat Code

Goal	Use
Compare categories	<code>sns.barplot()</code>
Show trend / time	<code>sns.lineplot()</code>
Number vs number	<code>sns.scatterplot()</code>
See a distribution	<code>sns.histplot()</code>
Spread / outliers	<code>sns.boxplot()</code>
See correlations	<code>sns.heatmap()</code>
Many relationships	<code>sns.pairplot()</code>

10. Mental Model

CATEGORY → **BAR**

TREND → **LINE**

NUMBER vs NUMBER → **SCATTER**

DISTRIBUTION → **HISTOGRAM**

SPREAD / OUTLIERS → **BOX**

CORRELATION → **HEATMAP**

MANY RELATIONSHIPS → **PAIRPLOT**

ML Reminder

You do not need to memorize every Seaborn parameter. Know **what plot to choose, what the axes mean, and how to interpret it**. That is what carries into ML/EDA.